DP Biology – C1.1

Enzymes and metabolism

- How do enzymes act on, and respond to, other molecules?
- Which parts of metabolism depend on one another?

SL statements C1.1.1–10. Textbook pp. 380–395.

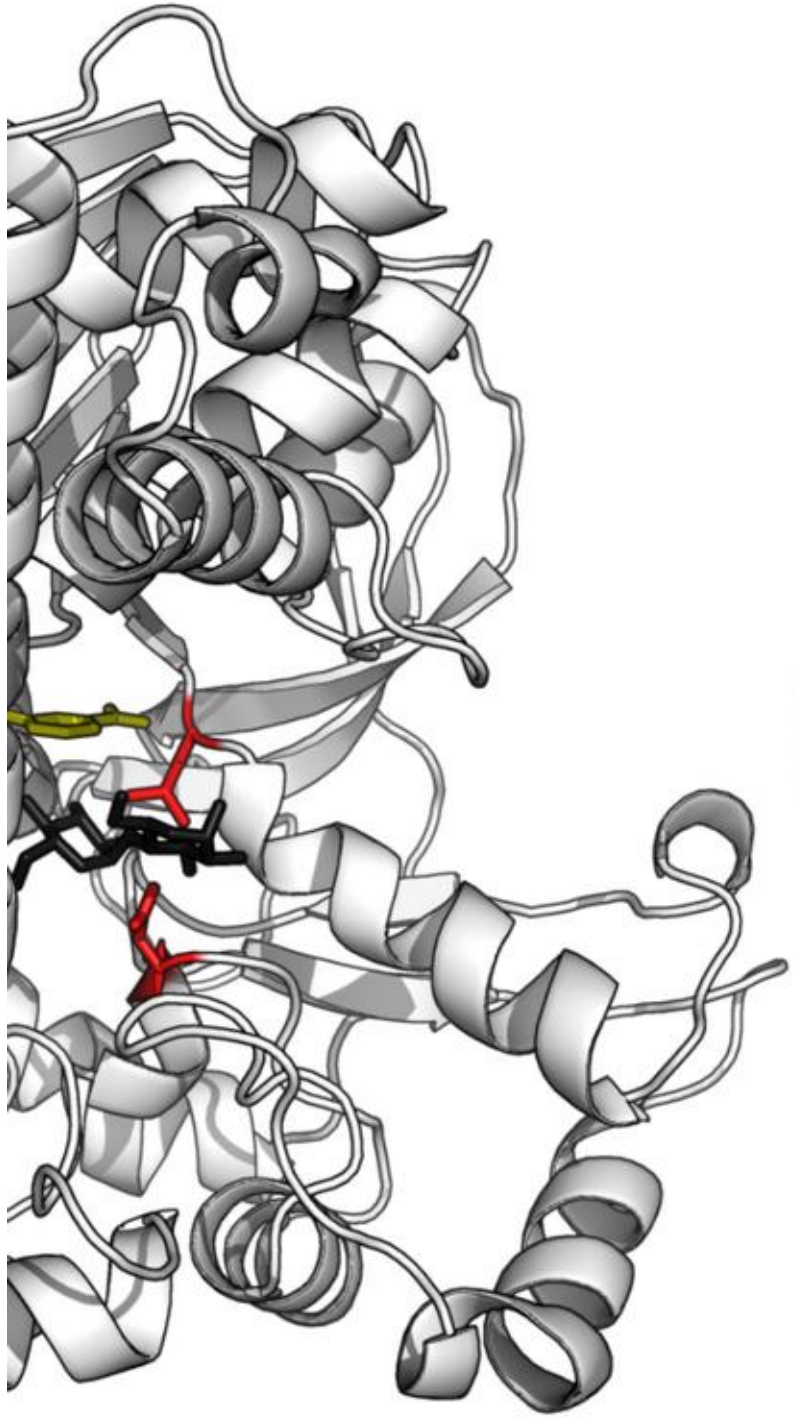


Image: model of a glucosidase enzyme. Thomas Shafee, CC BY 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.1.1–4	Explain why cells need enzymes, distinguish anabolic and catabolic reactions, and describe the active site
2	C1.1.5–7	Explain induced fit, collisions, immobilized enzymes, specificity and denaturation
3	C1.1.8–10	Explain how temperature, pH and substrate change the rate, calculate rates, and explain activation energy

How an enzyme works



Cells run thousands of reactions at once, each catalysed by its own **enzyme** whose **active site** fits its substrate.

As proteins, enzymes depend on **temperature** and **pH**; their rate depends on meeting substrate.

Big question

Why is a fever of 42 °C dangerous when 37 °C is fine?

Enzymes are catalysts

C1.1.1

A **catalyst** speeds a reaction without being used up. **Enzymes** are biological catalysts.

Without them, most cell reactions would be far too slow at body temperature. Enzymes also let the cell choose which reactions happen.

How much faster?

One enzyme speeds its reaction from a half-life of about 78 million years to a few milliseconds.

Think

Why can’t a cell just raise its temperature to speed up reactions?

Metabolism: a web of linked reactions

C1.1.2

Metabolism is all the reactions in a cell. Many are linked in **pathways**: each product is the next substrate, and each step has its own **specific** enzyme.



Why it matters

A cell can switch a whole pathway on or off by controlling one enzyme. One missing enzyme causes disease: in PKU, phenylalanine builds up.

Anabolic and catabolic reactions

C1.1.3

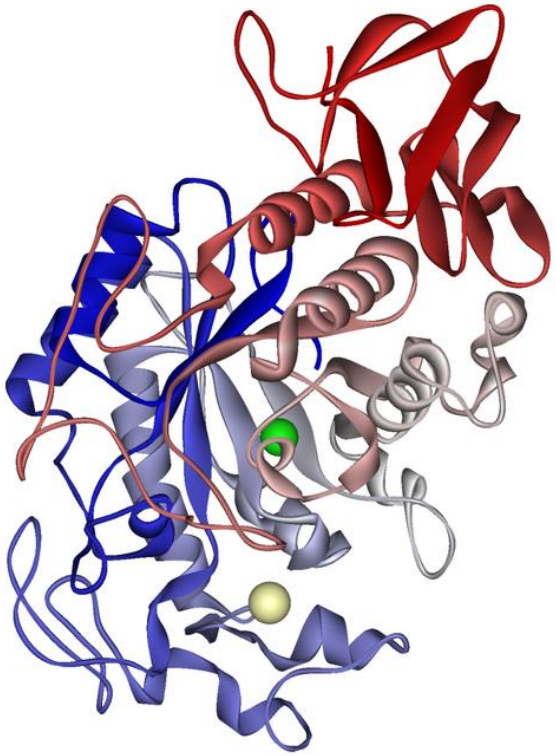
	Anabolic	Catabolic
What happens	Builds larger molecules	Breaks larger molecules down
Reaction	Usually condensation	Usually hydrolysis or oxidation
Energy	Needs energy (ATP)	Often releases energy
Examples	Protein synthesis, photosynthesis	Digestion, respiration

Memory aid

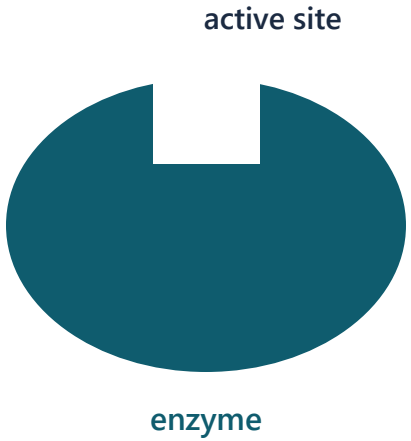
Anabolic builds; catabolic breaks down (a catastrophe).

Enzymes: folded proteins with an active site

C1.1.4



Salivary amylase, a globular protein. Public domain.





Enzymes are **globular proteins**. The **active site** is a pocket whose shape and chemistry are **complementary** to the substrate.

Common mistake

Active site and substrate are not the same shape; they are complementary.

Practice: classify and predict

C1.1.1–4

Classify

Digestion, protein synthesis, respiration, photosynthesis: anabolic or catabolic?

Pathway

E2 is blocked in A → B → C → D. What builds up?

Shape

A mutation changes the active site. Predict the effect.

Explain

Give a reason for each answer.

Check-in: class 1

C1.1.1–4

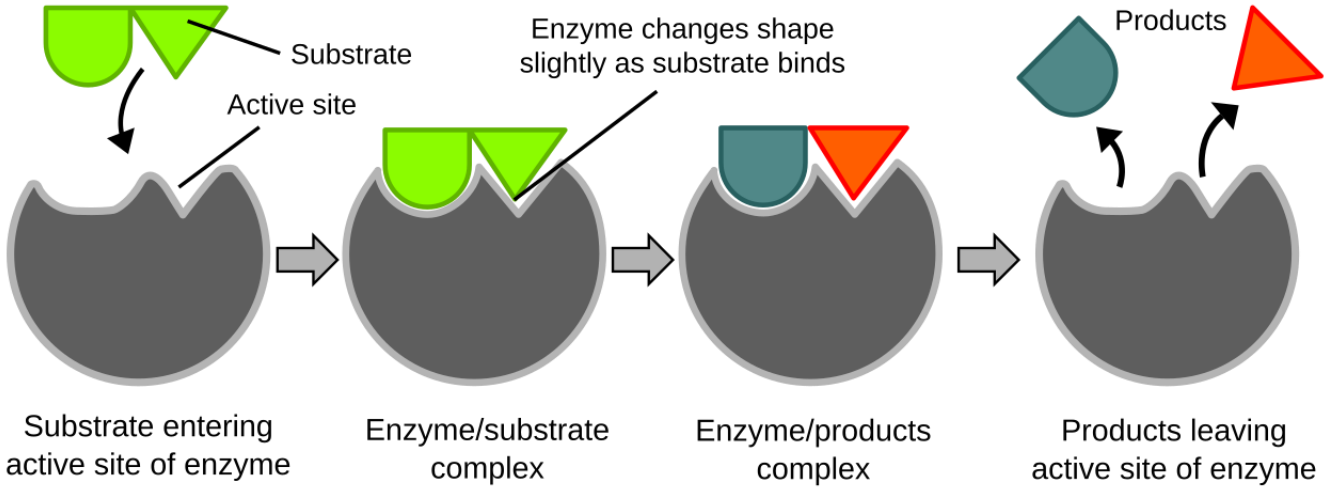
1. Define catalyst.
2. Explain why cells need enzymes.
3. Distinguish anabolic from catabolic reactions.
4. Define the active site.
5. Explain why a pathway needs many enzymes.

How to use this

Answer without your notes, then check against slides 3–8.

Induced-fit binding

C1.1.5



Induced fit. TimVickers, vectorized by Fvasconcellos, public domain.

As the substrate binds, **both enzyme and substrate change shape** to fit closely: **induced fit**. This strains bonds in the substrate so the reaction happens more easily.

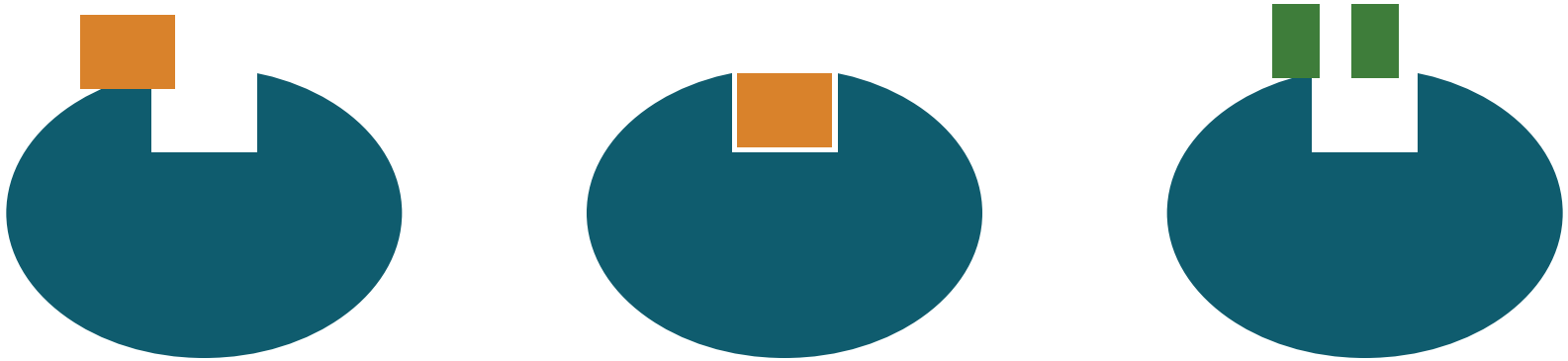
Compare

The older lock-and-key model pictured a rigid active site. Induced fit explains the flexibility better.

Steps of an enzyme-catalysed reaction

C1.1.5

Enzyme + substrate → **Enzyme-substrate complex** → Enzyme-product complex → Enzyme + products



drawn: substrate (orange) binds, reacts, products (green) leave

The enzyme is **unchanged** and is reused thousands of times a second.

Exam tip

Use the terms active site, enzyme-substrate complex and induced fit.

Molecular motion and collisions

C1.1.6

Substrate and active site must **collide**, in the right **orientation**. Molecules in solution move randomly, so collisions happen constantly.

Situation	What moves
Both dissolved	Both; small substrates move fastest
Enzyme fixed in a membrane	Only the substrate
Large insoluble substrate	Only the enzyme

Key idea

More frequent collisions (more substrate, higher temperature) mean a faster reaction.

Immobilized enzymes in industry

C1.1.6

Enzymes can be **immobilized** in beads or on surfaces. **Lactase** in beads makes lactose-free milk: milk flows past and lactose is hydrolysed.

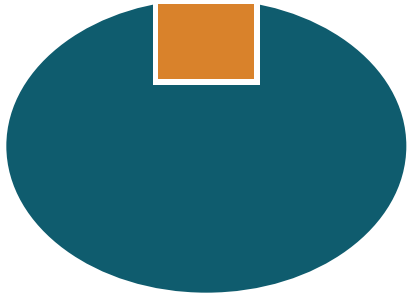
They can be reused, do not contaminate the product and are often more stable.

Think

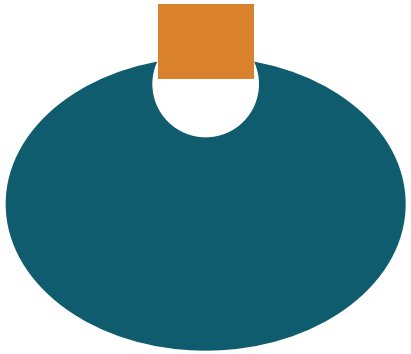
Many cell enzymes are fixed close together in membranes. Suggest one advantage for a pathway.

Specificity and denaturation

C1.1.7



fits: reaction



denatured: no fit

Specificity: only a substrate with a complementary shape and chemistry binds.

Denaturation: heat or extreme pH breaks the bonds holding the fold, the active site changes, and the substrate no longer fits. Usually permanent.

Link

Sequence decides fold; fold makes the active site (B1.2).

Practice: explain it

C1.1.5–7

Lactose-free

Why does milk from the lactase column contain no enzyme?

Specific

Why does amylase not digest proteins?

Boiled

Why does boiled enzyme no longer work when cooled?

Use

active site, complementary, denatured

Check-in: class 2

C1.1.5–7

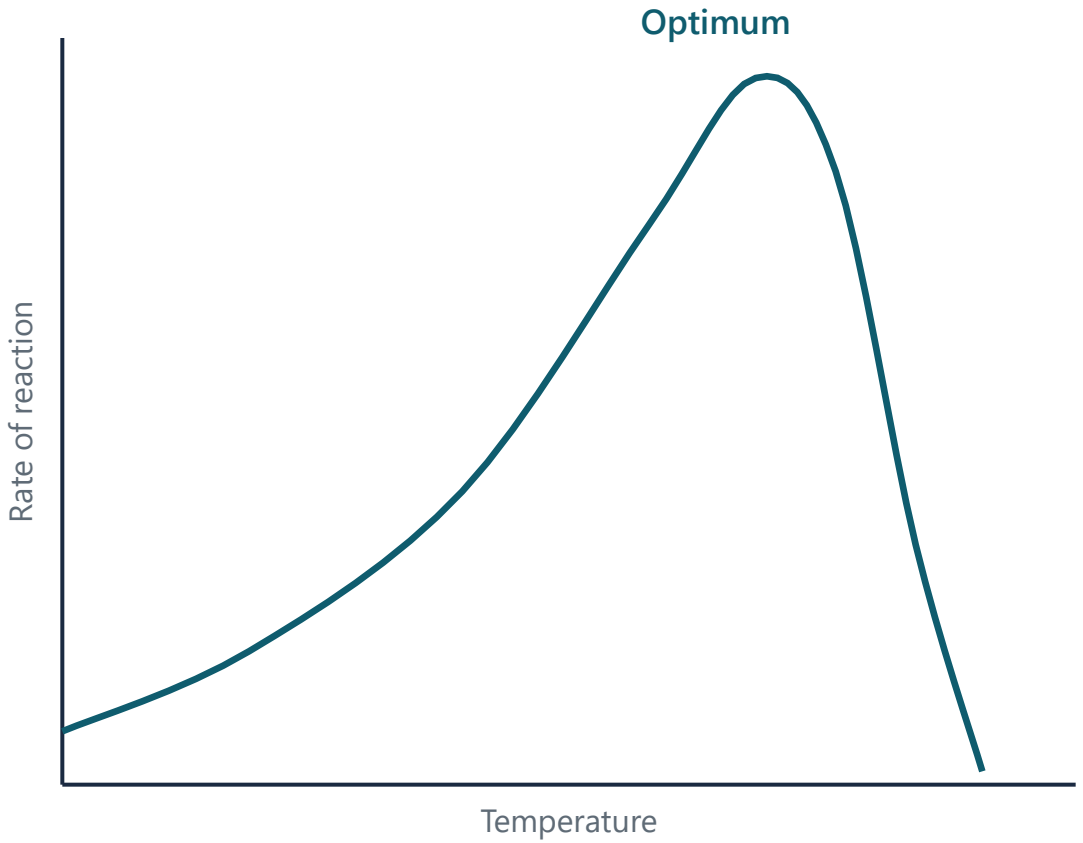
1. Outline the induced-fit model.
2. List the stages of an enzyme-catalysed reaction.
3. Explain why collisions matter.
4. Give two advantages of immobilized enzymes.
5. Explain specificity and denaturation using the active site.

How to use this

Answer without your notes, then check against slides 10–15.

Effect of temperature

C1.1.8



General shape only.

Rising: faster molecules, more frequent and energetic collisions: rate rises.

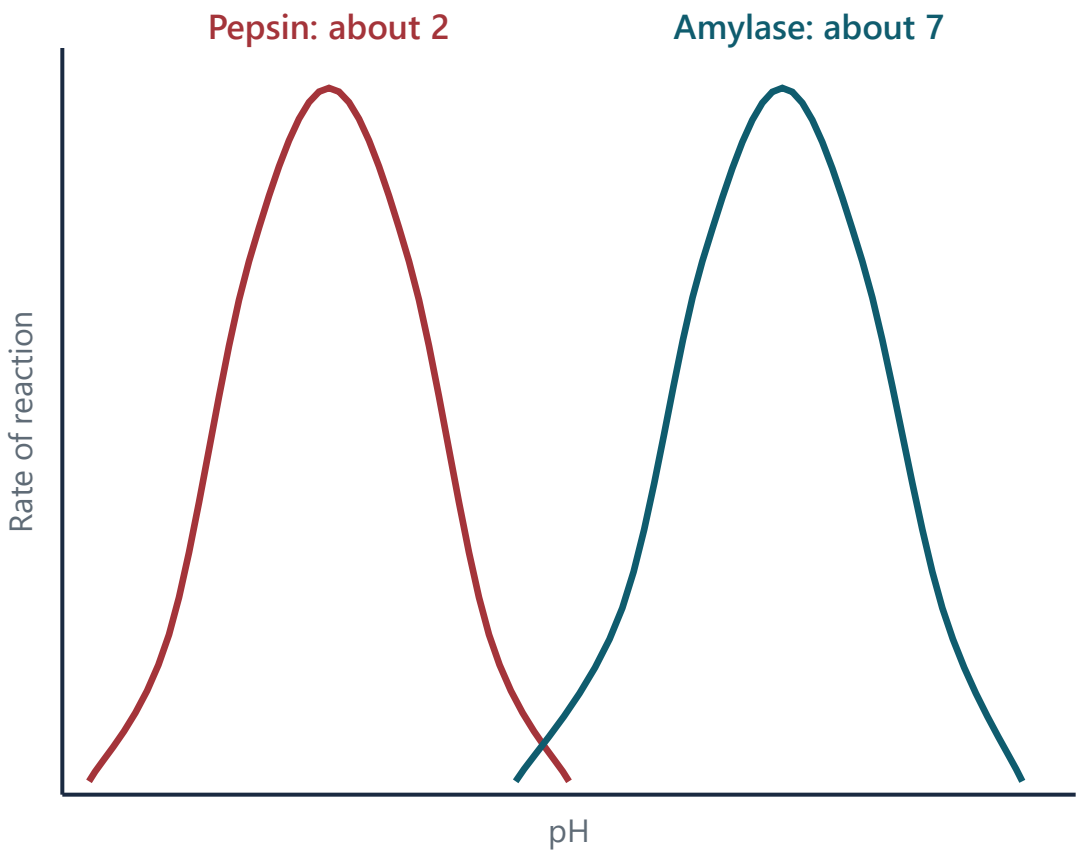
Above the optimum: bonds holding the shape break, the active site changes: rate falls fast, then stops.

Big question

At 42 °C vital enzymes begin to denature.

Effect of pH

C1.1.8



Each enzyme has an **optimum pH**, usually matching where it works.

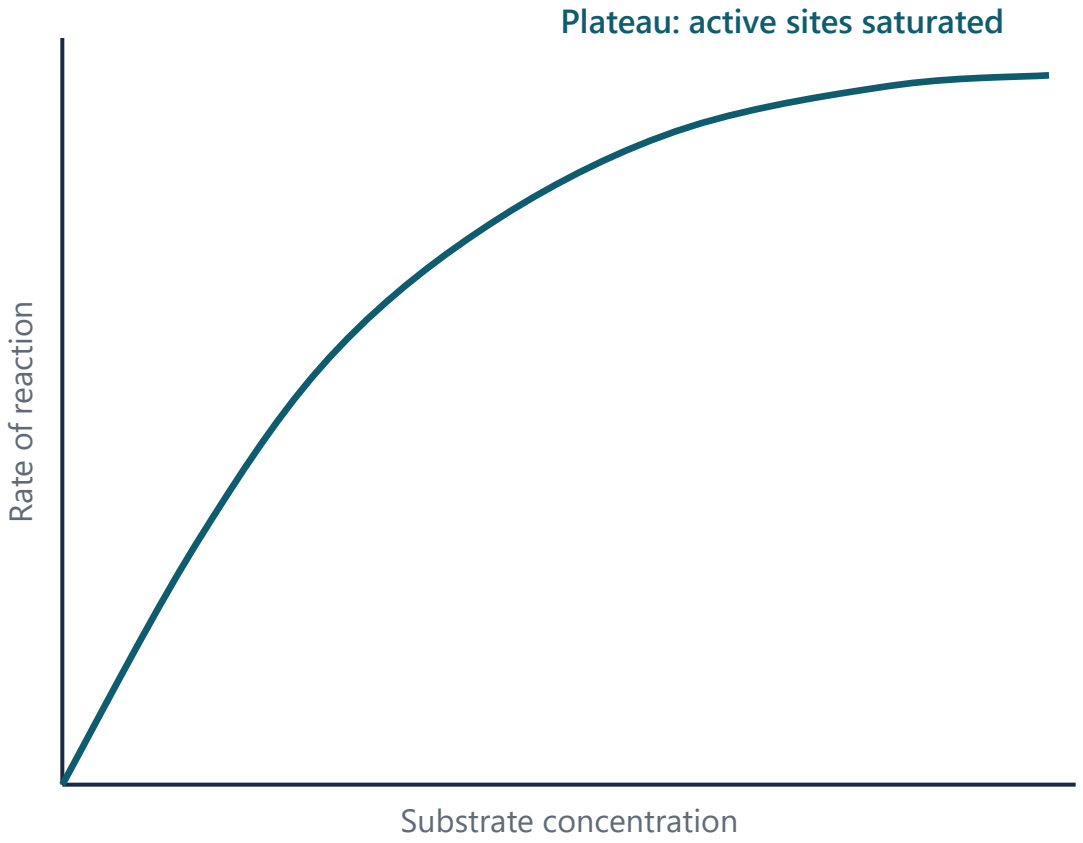
Away from it, H^+ changes charges on R-groups, so the active site changes and the rate falls.

Try it

Why does pepsin stop working in the small intestine?

Effect of substrate concentration

C1.1.8



More substrate, more collisions: the rate rises.

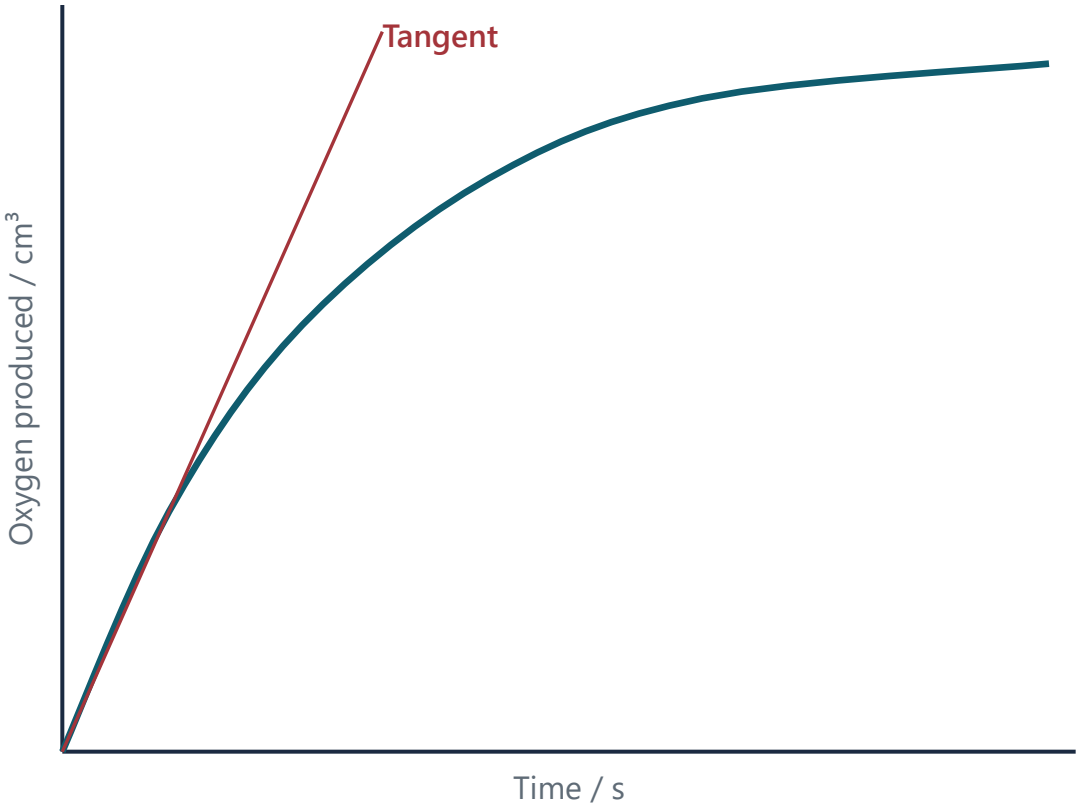
At high concentration every active site is busy: the rate **levels off**.

Common mistake

No optimum and no denaturation: the enzyme is working flat out.

Measuring the rate of a reaction

C1.1.9



Rate = product made per unit time. Compare the **initial rate**: the gradient at the start, when substrate is highest.

Worked example

Catalase made 6 cm^3 of oxygen in the first 20 s. Initial rate = 6 / 20 = 0.3 $cm^3\ s^{-1}$.

Designing an enzyme experiment

C1.1.8–9

Part	Example: temperature and catalase
Independent	Temperature: at least five values, water bath
Dependent	Oxygen collected in the first 30 s
Controlled	Volume and concentration of peroxide; amount of catalase; pH (buffer)
Repeats	Three at each temperature; calculate means

Safety
Eye protection: hydrogen peroxide is an irritant.

Working with data: catalase at three temperatures

C1.1.9

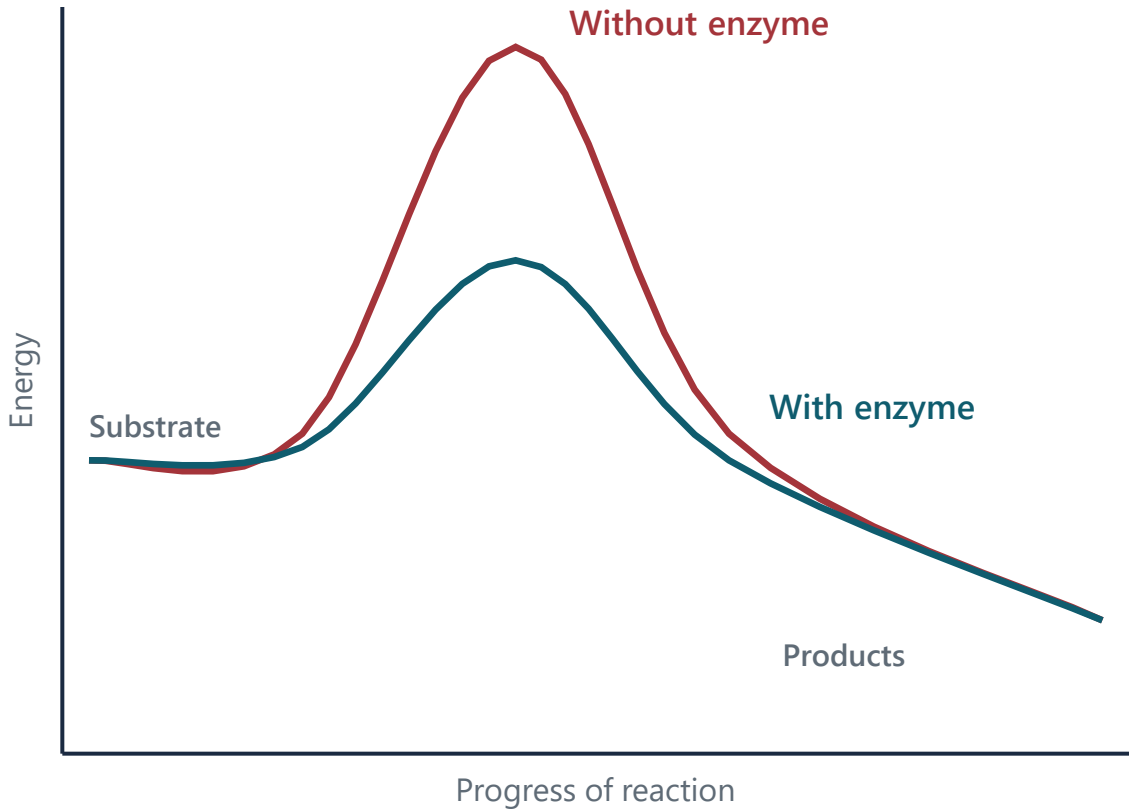
Oxygen (cm^3) released by the same amount of catalase. Invented values; realistic pattern.

Time / s	0	10	20	30	40	50
20 °C	0	1.2	2.3	3.3	4.2	5.0
35 °C	0	2.6	4.9	6.8	8.3	9.4
60 °C	0	0.6	0.9	1.1	1.2	1.2

1. Calculate each initial rate (first 10 s).
2. Compare and explain 20 and 35 °C.
3. Explain the 60 °C result.
4. Why does it level off at 60 °C?

Enzymes lower activation energy

C1.1.10



Activation energy is needed to start a reaction. An enzyme **lowers** it, so many more collisions succeed at body temperature.

The energy of substrate and products is unchanged.

Common mistake

Enzymes do not add energy; they lower the barrier.

Practice: sketch and explain

C1.1.8–10

Sketch

Rate against temperature, with the optimum marked.

Predict

Doubling enzyme at high substrate concentration.

Explain

Why a cool enzyme recovers but a boiled one does not.

Use

collisions, active site, denatured, activation energy

Check-in: class 3

C1.1.8–10

1. Explain why rate falls quickly above the optimum temperature.
2. Explain the plateau at high substrate concentration.
3. Explain how pH affects an enzyme.
4. State why initial rates are compared.
5. Explain how enzymes speed up reactions.

How to use this

Answer without your notes, then check against slides 17–24.

Common mistakes to avoid

Not this	But this
Enzymes are killed by heat.	They are not alive; they are denatured.
Active site and substrate are the same shape.	They are complementary.
High substrate denatures the enzyme.	Active sites are saturated.
Enzymes give energy to a reaction.	They lower activation energy.
Low temperature denatures enzymes.	It slows them; they work again when warmed.

Summary: enzymes and metabolism

C1.1.1–10

	Key idea
C1.1.1–3	Catalysts; metabolism; anabolic and catabolic
C1.1.4–5	Globular proteins; active site; induced fit
C1.1.6	Collisions; immobilized enzymes
C1.1.7	Specificity and denaturation
C1.1.8	Temperature, pH, substrate concentration
C1.1.9	Initial rates from data
C1.1.10	Lower activation energy

Check that you can...

1. **Explain** why cells need enzymes
2. **Distinguish** anabolic and catabolic reactions
3. **Explain** induced fit, specificity and denaturation
4. **Sketch** and **explain** how rate changes with temperature, pH or substrate
5. **Calculate** initial rates with units
6. **Explain** how enzymes lower activation energy

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Classify each as anabolic or catabolic: digestion, protein synthesis, respiration.
2. Explain why the rate of an enzyme-catalysed reaction falls at very high temperatures.
3. Sketch a graph showing how substrate concentration affects the rate, and label where the active sites are saturated.

Image credits

Glucosidase enzyme.png – Thomas Shafee – CC BY 4.0

Salivary alpha-amylase 1SMD.png – Own work. – Public domain

Induced fit diagram.svg – Created by TimVickers, vectorized by Fvasconcellos – Public domain

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